



Dodge being the tech support

Just purchased a device for someone? Here is a guide on how to avoid becoming their tech support. <https://digit.in/apr20-33>



Upgrade your home office

A list of things you will need to make the WFH experience better, which works just as well at the office. <https://digit.in/apr20-34>

Synology DiskStation DS1019+ NAS

Still top of the class

NAS devices are meant to last for a long period of time so it's natural to not see the lineup refreshing with the same gusto that one can see in, say the CPU or GPU segment. The Synology DS1019+ comes as a minor upgrade to the DS918+ which was one of the top NAS devices in the segment. How minor? Let's see.

Build Quality

The Synology DS1019+ is a 5-bay NAS, so right off the bat, you have one additional drive bay and with the DX517 expansion unit, you can get a total of 10 bays. Coming back to the DS1019+, the build is mostly the same as the internal chassis and the mainboard mount are the only metallic components while the rest of the unit is made up of plastic. The locking mechanisms and the HDD trays are still the same and there's no need to change things up here since everything works just fine.

The rear panel on which the ports and fans are mounted happen to be made of metal as well. We do see the main SoC mounted in a fashion that makes it slightly isolated from the chamber housing the hard drives and there's even spacing to ensure airflow over the SoC heatsink. Unlike the DS918+, the DS1019+ has both RAM slots populated and the modules are accessible once you remove the external plastic shroud. The OEM for the memory modules is Samsung using DDR3 chips carrying the label K4B4G1646E-BYMA indicating that they are rated for 1866 Mbps.

For the SoC, Synology is using the Intel Apollo Lake platform (Intel Atom J3455) which is incidentally what they used in the DS918+ as well. The Intel Atom J3455 is a 4 core / 4 thread



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processor with Intel HD 500 Graphics featuring 12 Execution Units. It has a base clock of 1.5 GHz and can rise to 2.3 GHz. Synology could have upgraded to the Gemini Lake Intel Atom J4105 for no additional cost but given that NAS devices are meant for 24x7 operation with very little bursty workloads, we don't think it would have helped much from a compute perspective. However, Gemini Lake has support for DDR4 RAM, USB 3.1 and USB Type-C interfaces. This would certainly have helped enthusiasts who are more likely to go for a high-end NAS such as the DS1019+.

So with the DS1019+, we have the same CPU as the DS918+, the same memory capacity, and similar interface options. It comes with more memory pre-installed and one additional bay which is why we're looking at it as a minor upgrade to the DS918+. Most of the value addition with Synology is always with their OS, DiskStation Manager.

Performance

Synology supplied us with just two hard drives which were insufficient for our benchmarks so we threw in an NVMe SSD for caching and switched out the hard drives that Synology gave us to incorporate a RAID 0 and RAID 6 setup with 4x WD Red NAS Drives. The DS918+ was one of the best

performing NAS devices we'd tested two years ago so we had high hopes for the DS1019+.

In our RAID 0 benchmarks, the Synology DS1019+ outperforms the DS918+ by a narrow margin.

The 4K read throughput goes a little over 32K IOPS whereas the write throughput is just about over 25K IOPS. Compared to the DS918+, this is about 500-700 IOPS greater on both benchmarks. We also ran the tests with and without the SSD caches to see a huge difference in latency. Without the SSD Cache, reading from the NAS from an idle state would often take 10-14 seconds whereas with the SSD cache this came down to less than 200 ms. The SSD cache certainly helps in making the NAS a lot more responsive than NAS devices without it. The benefits of this can be seen when you've stored a lot of photos on your drives and you quickly want to skim through a massive library to get the few files you want. It should also be noted that with DSM 7.0, Synology revamped the photo library app in the form of Synology Photos and the cumulative benefit of the improved app and SSD caching is what we've observed.

Verdict

The Synology DS1019+ is a minor upgrade to the DS918+. In fact, we'd say that it replaces the DS918+ given that Indian pricing for the two devices are just about the same. The additional hard drive bay and RAM certainly help with the value addition and DSM has improved by leaps and bounds.

—Mithun Mohandas

SPECIFICATIONS

CPU: Intel Celeron J3455 Quad Core | RAM: 8 GB DDR3L-1866 | BAYS: 5 (+2 slots for M.2 SSDs) | HOT SWAP SUPPORT: Yes | I/O: eSATA, 2x USB 3.0 | LAN PORTS: 2x Gigabit | WEIGHT: 2.54 kg | Fan: 2x 92 mm | WARRANTY: 3 years.

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